I. Single choice questions

1. A continuous-time signal is sampled at fs=400Hz, yielding a sequence of 200 samples. After performing a 400-point DFT on this 200-sample sequence, the true physical frequency interval between two adjacent spectral lines in the frequency domain is (　　)Hz.

A. π/100　　B. π/200　　C. 1　　D. 2

2. The discrete-time system $y[n]=x[n](u[n]-u[n-5])$ is (　　).

A. Linear time-varying　　B. Nonlinear time-varying

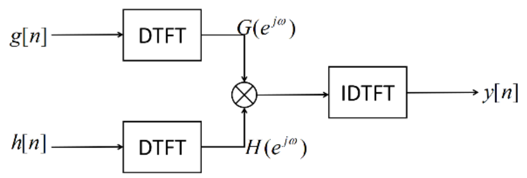
C. Linear time-invariant　　D. Nonlinear time-invariant

3. For a complex sequence x[n] of length N, its N-point DFT is X[k]. Regarding the symmetry of X[k], which of the following is **incorrect**?(　　).

A. $x^*[n] \overset{DFT}{\longleftrightarrow} X^*[\langle -k \rangle_N]$　　B. $\mathrm{x}_{ca}[n] \overset{DFT}{\longleftrightarrow} jX_{im}[k]$

C. $\mathrm{x}_{im}[n] \overset{DFT}{\longleftrightarrow} X_{ca}[k]$　　D. $\mathrm{x}_{re}[n] \overset{DFT}{\longleftrightarrow} X_{cs}[k]$

4. As shown in the figure, which of the following correctly describes the relationship among $g[n]$, $h[n]$, and $y[n]$? (　　).



A. $y[n]$ is the linear convolution of $h[n]$ and $g[n]$

B. $y[n]$ is the circular convolution of $h[n]$ and $g[n]$

C. $y[n]=G(e^{j\omega})H(e^{j\omega})$　　D. $y[n]=g[n]h[n]$

5. Given that the region of convergence for the Z-transform of a sequence is $|Z|>0.5$, the sequence is (　　).

A. a finite-length sequence　　B. a right-sided sequence

C. a left-sided sequence D. a two-sided sequence

6. Before sampling an analog signal, the signal must pass through an anti-aliasing filter. This filter is ( ).

A. a lowpass filter B. a highpass filter

C. a multiband filter D. a bandstop filter

7. Regarding the design of IIR filters using the bilinear transformation, which of the following statements is **correct**? ( ).

A. The transformation used is a multi-valued mapping from the s-plane to the z-plane.

B. It is not recommended for designing bandpass and bandstop filters.

C. There is a linear relationship between digital frequency and analog frequency.

D. It maps a stable analog filter to a stable digital filter.

8. The inverse system of a causal linear-phase FIR filter is ( ).

A．a causal stable system B．a minimum-phase system

C．a maximum-phase system D．not a causal stable system

9. When designing an IIR filter using the bilinear transformation, the relationship equation between the analog angular frequency Ω and the digital angular frequency ω is ( ).

A. $\Omega = \frac{1}{T}\tan\left(\frac{\omega}{2}\right)$ B. $\Omega = \frac{2}{T}\tan\left(\frac{\omega}{2}\right)$

C. $\Omega = \frac{1}{T}\tan(\omega)$ D. $\Omega = \frac{2}{T}\tan(\omega)$

10. For several common analog prototype filters of the same order, the relationship between their transition bandwidths is ( ).

A. Butterworth > Elliptic > Chebyshev B. Chebyshev > Butterworth > Elliptic

C. Elliptic > Butterworth > Chebyshev D. Butterworth > Chebyshev > Elliptic

II. Blank-fill questions

1. For a discrete-time LTI system describe by the difference equation

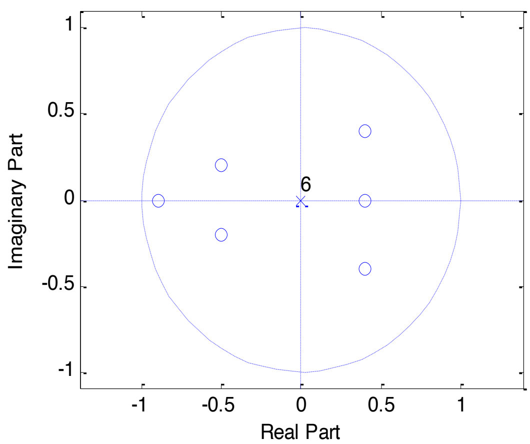
$y[n]-y[n-1]=\frac{1}{8}(x[n]-x[n-4])$, the unit impulse response of this LTI system is $\{h[n]\}=$__________

2. Given a complex sequence v[n]=x[n] + jy[n] of length 4 with its 4-point DFT：V[0]=-2+j6, V[1]=3+j5, V[2]=6+j4, V[3]=-1+j8. Without computing the IDFT of V[k], determine the 4-point DFTs of the real sequence x[n] and y[n]:

X[k]=__________Y[k]=_______________。

3. IIR filter $y[n]=\sum_{k=0}^{M} p_k x[n-k]-\sum_{l=0}^{N} d_l y[n-l]$, which $M>N$, is realized by canonic structure, so the number of delay is_______

4. A causal LTI system is as shown in the following pole-zero plot, so this system is a _______(FIR or IIR) filter, and it is a ________(linear phase, minimum phase or maximum phase) system.



5. The computational complexity of the complex addition operations in an N-point sequence FFT is _________

6. A 8 points average filter is described by $y[n]=\frac{1}{8}\sum_{l=0}^{7} x[n-l]$, please find out its group delay $\tau(w)$=_______

7. $x[n]$is a causal sequence, its Z transform $X(z)=\frac{0.9-z^{-1}}{1-0.9z^{-1}}$, so $x[1]$=____

8. When designing an IIR digital high-pass/band-pass/band-stop filter using the bilinear transform method, after obtaining the analog low-pass filter, one may either apply analog spectral inversion or use ____________to obtain the target IIR digital filter.

III. Calculations

1. Given a signal $x(t)=\cos\left(\frac{3\pi}{2}t\right)+\cos\left(\frac{39\pi}{8}t\right)+\sin(3\pi t)$

sampled at $f_s=3\text{Hz}$,

1）Write the discrete-time sequence $x[n]$. Determine whether this signal is periodic. If yes, find its period.

2）Can the continuous-time signal $x(t)$ be recovered from $x[n]$? Explain your reasoning.

3）Write out the 16-point DFT of $x[n], 0\le N\le 15$

2. The autocorrelation sequence $c[n]$ of $x[n]$ is defined as

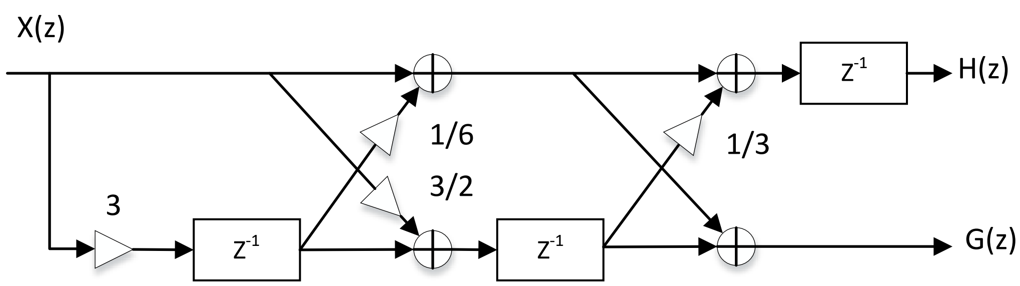
$$c[n]=\sum_{k=-\infty}^{+\infty}x(k)x(k-n)$$

Given that the Z-transform of $x[n]$ is $X(z)$, express the Z-transform of $c[n]$ in terms of $X(z)$.

3. A filter has the following structure.

1) Determine its $H(z)$ and $G(z)$.

2) For H(z), determine whether it is a canonical structure.



4. Find the system functions $H_1(z)$, $H_2(z)$, $H_3(z)$ and $H_4(z)$ for the four types of real-coefficient linear-phase FIR filters with the lowest order that has a zero at z_1=0.3+j0.4.

5. The signal y[n] consists of an original signal x[n] and two echoes：

$$y[n] = x[n] + 0.5x[n - n_d] + 0.25x[n - 2n_d]$$

Design a filter to recover x[n] from y[n], and determine its stability.

6. The specification for a multiband filter are given as:

$$\omega_{s1} = 0.08\pi \text{ ! } \omega_{p1} = 0.12\pi \text{ ! } \omega_{p2} = 0.28\pi \text{ ! } \omega_{s2} = 0.32\pi ,$$

$$\omega_{s3} = 0.58\pi \text{ ! } \omega_{p3} = 0.62\pi \text{ ! } \omega_{p4} = 0.76\pi \text{ ! } \omega_{s4} = 0.84\pi ,$$

with all passband and stopband ripples constrained to be less than 0.01. Using the window method and selecting an appropriate window from the Appendix, design a linear-phase FIR filter with minimum length that satisfies these specifications.

Appendix： Table1 Some fixed window functions

Rectangular：	$w[n] = \begin{cases} 1 & 0 \le \lvert n \rvert \le M \\ 0 & otherwise \end{cases}$
hann：	$w[n] = \frac{1}{2}[1 + \cos(\frac{2\pi n}{2M+1})], -M \le n \le M$
hamming：	$w[n] = 0.54 + 0.46\cos(\frac{2\pi n}{2M+1}), -M \le n \le M$
blackman：	$w[n] = 0.42 + 0.5\cos(\frac{2\pi n}{2M+1}) + 0.08\cos(\frac{4\pi n}{2M+1}), -M \le n \le M$

Table 2 Properties of some fixed window functions

Type of window	Main lobe width Δ_{ML}	Relative Sidelobe Level $A_{s\ell}$	Minimum Stopband Attenuation	Transition Bandwidth $\Delta\omega$
Rectangular	4π/(2M+1)	13.3dB	20.9dB	0.92π/M
Hann	8π/(2M+1)	31.5dB	43.9dB	3.11π/M
Hamming	8π/(2M+1)	42.7dB	54.5dB	3.32π/M
Blackman	12π/(2M+1)	58.1dB	75.3dB	5.56π/M